CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · FORMULA BANK

Chapter 6: Sets

Every formula and result of the chapter on one sheet

1	Membership	$a \in A, a \notin A$
2	Cardinal number	$n(A)$
3	Empty set	ϕ or $\{\}, n(\phi) = 0$
4	Number of subsets	2^n
5	Number of proper subsets	$2^n - 1$
6	Union	$A \cup B$: in A or B
7	Intersection	$A \cap B$: in both
8	Difference	$A - B$: in A, not in B
9	Complement	$A' = \xi - A$
10	Counting formula	$n(A \cup B) = n(A) + n(B) - n(A \cap B)$
11	Only A	$n(A - B) = n(A) - n(A \cap B)$
12	Complement count	$n(A') = n(\xi) - n(A)$
13	De Morgan's laws	$(A \cup B)' = A' \cap B', (A \cap B)' = A' \cup B'$